HOMEWORK 8

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4.1. Suppose the permutation σ takes $(1,2,3,4,5)$ to $(5,4,1,2,3)$.

(a) Find sign of σ;
(b) What does $\sigma^2 := \sigma \circ \sigma$ do to $(1,2,3,4,5)$?
(c) What does the inverse permutation σ^{-1} do to $(1,2,3,4,5)$?
(d) What is the sign of σ^{-1}?

Solution. (a) $(5,4,1,2,3)$ has $4+3=7$ inversions, so sign $\sigma = -1$. For (b), you simply apply the rule twice. It will give $(3,2,5,4,1)$. For (c), it inverts the rules. $\sigma(1) = 5 \implies 1 = \sigma^{-1}(5)$, and so on, giving us $(3,4,5,2,1)$. For (d), $(3,4,5,2,1)$ has $2+2+2+1=7$ inversions, so sign $\sigma^{-1} = -1$. ■

4.2. Let P be a *permutation matrix*, i.e. an $n \times n$ matrix consisting of zeroes and ones and such that there is exactly one 1 in every row and every column.

(a) Can you describe the corresponding linear transformation? That will explain the name.
(b) Show that P is invertible. Can you describe P^{-1}?
(c) Show that for some $N > 0$

$$P^N := \underbrace{PP\dots P}_{N \text{ times}} = I.$$

Use the fact that there are only finitely many permutations.

Solution. (a). The standard basis is permuted: P sends each e_j to some e_k, so it rearranges the coordinates of a vector. ■

Proof. (b). There is exactly one in every row and every column. Let $\alpha = \pm 1$ keep track of the column swapping, since each swap flips the sign. If $a_{11} = 0$, we may column swap so that $a_{11} = 1$. We can proceed like this inductively for $a_{22}, a_{33}, \dots, a_{nn}$, giving us I. Hence, we get $\det(P) = \alpha \det(I) \neq 0$. Therefore, P is invertible. Moreover, $P^{-1} = P^T$: the columns of P are distinct standard basis vectors, so the dot product of column i and column j is 1 if $i = j$ and 0 otherwise, giving $P^T P = I$. ■

Proof. (c). Each product of permutation matrices is itself a permutation matrix. Each P maps the standard basis to a standard basis vector. P^c is invertible by $\det(P \cdots P) = \det(P) \dots \det(P) \neq 0$ by part b. Since there are finitely many permutation matrices, given by $n!$, $P^1, P^2, P^3, \dots$ cannot all be distinct. Therefore, for some $a < b$, $P^a = P^b$. By invertibility, multiplying by P^{-a} gives $I = P^{b-a}$. Let $N = b - a > 0$. Hence, we have completed the proof. ■

4.3. Why is there an even number of permutations of $(1, 2, \ldots, 9)$ and why are exactly half of them odd permutations?
Hint: This problem can be hard to solve in terms of permutations, but there is a very simple solution using determinants.

Proof. There are 9! permutations of [9]. Since 9! has a factor of 2 inside of it, it is even.

Let $A \in \mathcal{M}_{9\times 9}$, where each entry is 1. $\det A = \sum_{\sigma\in \mathrm{Perm}(9)} \operatorname{sgn}(\sigma) \prod_{i=1}^{9} a_{\sigma(i),i}$. Since all entries are one, each product is 1, so this will result in $\det(A)$ = #evens - #odds permutations. Since all rows of A are equal, $\det(A) = 0$. So # even - # odd $= 0 \implies$ # even = # odd. ∎

4.4. If σ is an odd permutation, explain why σ^2 is even but σ^{-1} is odd.

Solution. This is because it takes $2k + 1$ column swaps for a single σ. When squaring, you get an even amount of swaps, hence making sign $\sigma = -1$ and sign $\sigma^2 = 1$. For σ^{-1}, you are still performing the same amount of swaps, since it is merely an inverse, swapping the rules. So, sign $\sigma^{-1} = -1$. ∎

4.5. How many multiplications and additions is required to compute the determinant using formal definition (4.2) of the determinant of an $n \times n$ matrix? Do not count the operations needed to compute $\operatorname{sign}\sigma$.

Solution. This requires $(n-1)n!$ multiplications. We multiply n numbers, so we get $n-1$ multiplications, and since we are dealing with permutations, this can be done in $n!$ ways. So, $(n-1)n!$ is the answer. For addition, we have $n!$ terms, and adding each one up takes $n! - 1$ addition symbols. Therefore, it takes $n! - 1$ additions. ∎

5.1. Evaluate the determinants using any method

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & -5 \\ 6 & -4 & 3 \end{vmatrix}, \qquad \begin{vmatrix} 1 & -2 & 3 & -12 \\ -5 & 12 & -14 & 19 \\ -9 & 22 & -20 & 31 \\ -4 & 9 & -14 & 15 \end{vmatrix}.$$

Solution. Using the cofactor method for 3×3: $-\begin{vmatrix} 1 & -5 \\ 6 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 6 & -4 \end{vmatrix} = -49$.
For the 4×4 : Using $R_2 + 5R_1$, $R_3 + 9R_1$, $R_4 + 4R_1$ and expanding along the first column,

$$\det = \begin{vmatrix} 2 & 1 & -41 \\ 4 & 7 & -77 \\ 1 & -2 & -33 \end{vmatrix} = -100$$

by cofactor expansion. ∎

5.2. Use row (column) expansion to evaluate the determinants. Note, that you don't need to use the first row (column): picking row (column) with many zeroes will simplify your

calculations.

$$\begin{vmatrix} 1 & 2 & 0 \\ 1 & 1 & 5 \\ 1 & -3 & 0 \end{vmatrix}, \qquad \begin{vmatrix} 4 & -6 & -4 & 4 \\ 2 & 1 & 0 & 0 \\ 0 & -3 & 1 & 3 \\ -2 & 2 & -3 & -5 \end{vmatrix}.$$

Solution. Expanding along column 3, $-5\begin{vmatrix} 1 & 2 \\ 1 & -3 \end{vmatrix} = 25$. For the second matrix, row swap R_2 and R_1. Then, we get $-\begin{vmatrix} 2 & 1 & 0 & 0 \\ 4 & -6 & -4 & 4 \\ 0 & -3 & 1 & 3 \\ -2 & 2 & -3 & -5 \end{vmatrix} = -2\begin{vmatrix} -6 & -4 & 4 \\ -3 & 1 & 3 \\ 2 & -3 & -5 \end{vmatrix} + \begin{vmatrix} 4 & -4 & 4 \\ 0 & 1 & 3 \\ -2 & -3 & -5 \end{vmatrix} =$ -32. ■

5.3. For the $n \times n$ matrix

$$A = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & a_0 \\ -1 & 0 & 0 & \dots & 0 & a_1 \\ 0 & -1 & 0 & \dots & 0 & a_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & a_{n-2} \\ 0 & 0 & 0 & \dots & -1 & a_{n-1} \end{pmatrix}$$

compute $\det(A + tI)$, where I is $n \times n$ identity matrix. You should get a nice expression involving $a_0, a_1, \dots, a_{n-1}$ and t. Row expansion and induction is probably the best way to go.

Solution. We claim $\det(A+tI) = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0$ and prove it by induction on n. For $n = 1$, $\det(t + a_0) = t + a_0$. For $n > 1$, the first row of $A + tI$ is $(t, 0, \dots, 0, a_0)$. Deleting row 1 and column 1 leaves a similar matrix of size $n - 1$ with $a_1, \dots, a_{n-1}$ on the furthest right column, whose determinant is $t^{n-1} + a_{n-1}t^{n-2} + \cdots + a_1$ by induction. Deleting row 1 and column n leaves an upper triangular matrix with -1 on the diagonal, whose determinant is $(-1)^{n-1}$. Therefore, we get

$$\det(A+tI) = t(t^{n-1} + a_{n-1}t^{n-2} + \cdots + a_1) + (-1)^{1+n}(-1)^{n-1}a_0 = t^n + a_{n-1}t^{n-1} + \cdots + a_1 t + a_0.$$

■

5.4. Using cofactor formula compute inverses of the matrices

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 19 & -17 \\ 3 & -2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 3 & 5 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

Solution. For 2×2 matrices we have $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, so

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}^{-1} = -\frac{1}{2}\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}, \begin{pmatrix} 19 & -17 \\ 3 & -2 \end{pmatrix}^{-1} = \frac{1}{13}\begin{pmatrix} -2 & 17 \\ -3 & 19 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 3 & 5 \end{pmatrix}^{-1} = \frac{1}{5}\begin{pmatrix} 5 & 0 \\ -3 & 1 \end{pmatrix}.$$

For the 3×3 matrix, expanding along the first row gives $\det = 1(1-2) - 1(2-0) = -3$. Computing each cofactor,

$$C = \begin{pmatrix} -1 & -2 & 2 \\ -1 & 1 & -1 \\ 2 & -2 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}^{-1} = -\frac{1}{3}C^T = \frac{1}{3}\begin{pmatrix} 1 & 1 & -2 \\ 2 & -1 & 2 \\ -2 & 1 & 1 \end{pmatrix}.$$

■

5.5. Let D_n be the determinant of the $n \times n$ tridiagonal matrix

$$\begin{pmatrix} 1 & -1 & \cdots & \cdots & 0 \\ 1 & 1 & -1 & \cdots & \vdots \\ \vdots & 1 & \ddots & \ddots & \vdots \\ \vdots & \cdots & \ddots & 1 & -1 \\ 0 & \cdots & \cdots & 1 & 1 \end{pmatrix}.$$

Using cofactor expansion show that $D_n = D_{n-1} + D_{n-2}$. This yields that the sequence D_n is the Fibonacci sequence $1, 2, 3, 5, 8, 13, 21, \ldots$

Solution. Let A be the matrix above. Using cofactor expansion along the first row, we get A_{11} = same matrix, just with $n-1$ dimension. This is D_{n-1}. For A_{12}, we get $A_{12} = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & -1 & & \\ 0 & 1 & 1 & \ddots & \\ \vdots & & \ddots & \ddots & -1 \\ 0 & & & 1 & 1 \end{pmatrix}$. Expanding along the first column, we get the same matrix once again, but this time with $n-2$ dimension. This is D_{n-2}. Therefore, $\det(A) = (1)\det(A_{11}) - (-1)\det(A_{12}) = \det(A_{11}) + \det(A_{12})$, so $D_n = D_{n-1} + D_{n-2}$. Since $D_1 = 1$ and $D_2 = 1+1 = 2$, this is the Fibonacci sequence. ■